

INTERNSHIP PROJECT REPORT

Student Performance Prediction Using Machine Learning

A beginner-level machine learning classification project using Python and Scikit-learn.

1. Introduction

Machine Learning allows computers to learn patterns from data and use those patterns to make predictions. This project applies a supervised machine learning technique to student academic data.

2. Problem Statement

The aim is to predict whether a student is likely to pass or fail using study hours, attendance, assignments completed, and previous exam score.

3. Objectives

The main objectives are to understand a basic ML workflow, prepare data, visualize relationships, train a classification model, evaluate its performance, and generate a prediction for a new student.

4. Technologies Used

Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, and Scikit-learn.

5. Dataset

The project uses a reproducible educational sample dataset containing 300 records. Features include Study_Hours, Attendance, Assignments_Completed, and Previous_Exam_Score. The target variable is Pass, represented by 0 for Fail and 1 for Pass.

6. Methodology

The dataset is loaded and checked for missing values. Features and target are separated. The data is split into 80% training and 20% testing sets. StandardScaler is applied to numerical features. Logistic Regression is then trained on the training data.

7. Model Used

Logistic Regression is suitable for this project because the target has two classes: Pass and Fail. It estimates the probability of belonging to each class.

8. Results

The trained model achieved an accuracy of 81.67% on the test data. The confusion matrix and classification report are included in the Jupyter Notebook.

9. Sample Prediction

For a sample student with 7 study hours, 85% attendance, 8 completed assignments, and a previous exam score of 72, the trained model produces a Pass/Fail prediction along with class probabilities.

10. Conclusion

The project demonstrates the complete machine learning workflow from data preparation to prediction. It provides a simple example of how classification can be applied to academic data.

11. Future Scope

Future versions can use a larger real-world dataset, compare Logistic Regression with Decision Tree, Random Forest, and other algorithms, improve feature engineering, and deploy the model through a web application.

12. Learning Outcomes

Through this project, a learner can practice data handling with Pandas, visualization with Matplotlib, preprocessing with StandardScaler, classification with Logistic Regression, and evaluation using accuracy, classification report, and confusion matrix.

Project Files

The accompanying Jupyter Notebook contains the complete Python implementation, model training, evaluation, visualization, and sample prediction.

